

SHARPIE: STREAMLINING HUMAN-AI REINFORCEMENT LEARNING EXPERIMENTS

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1. What is SHARPIE?

RL is being studied in increasingly complex

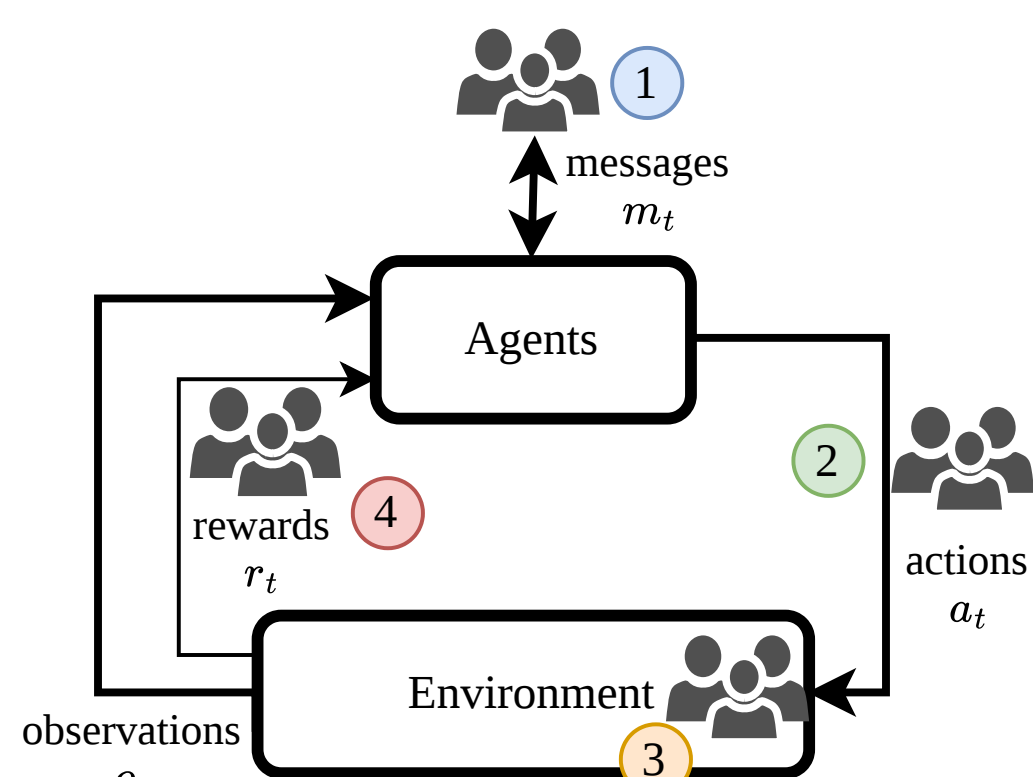
multi-human multi-agent settings

with humans at the center & rich interaction paradigms.

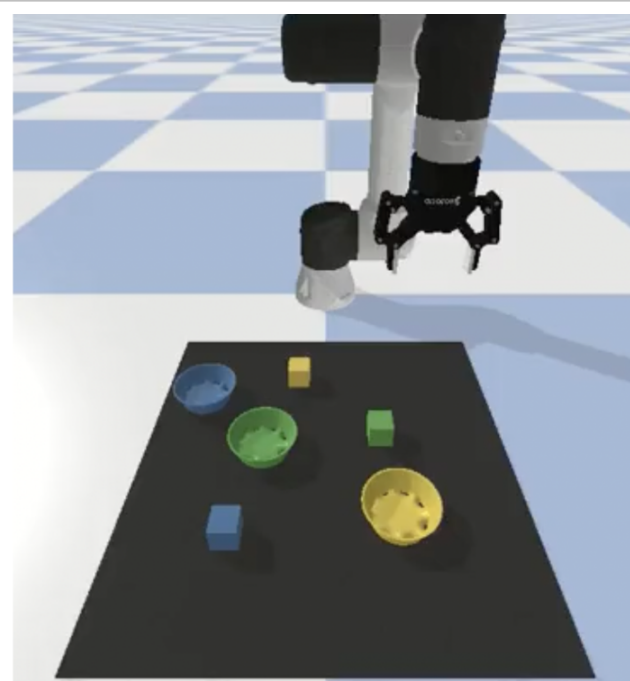
Implementing **human-RL experiments** presents specific **infrastructural challenges**. SHARPIE

Shared Human AI Reinforcement learning
Platform
for Interactive Experiments.

is a web-based human-RL experimentation platform tackling these challenges.



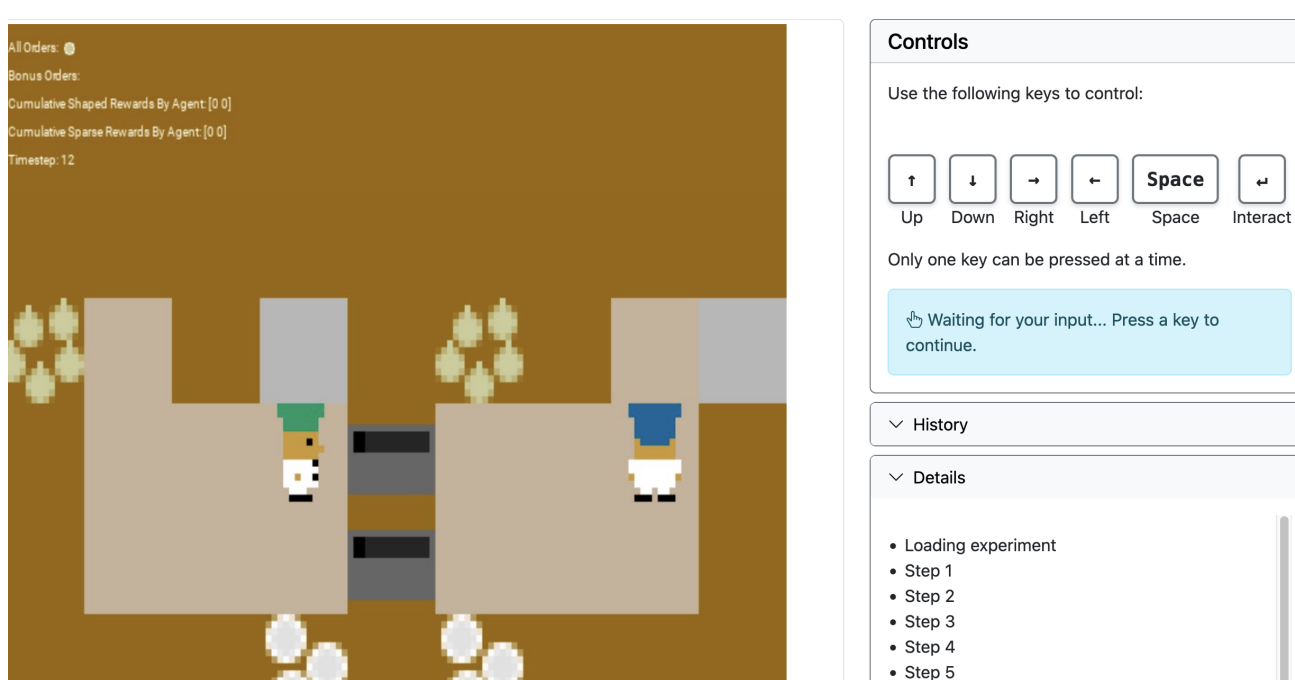
2. Is SHARPIE easy to use?



(a) Interactions with messages ① with SayCan in tabletop: 17 LoC



(b) Interactions with actions ② with Dagger in Mario: 65 LoC



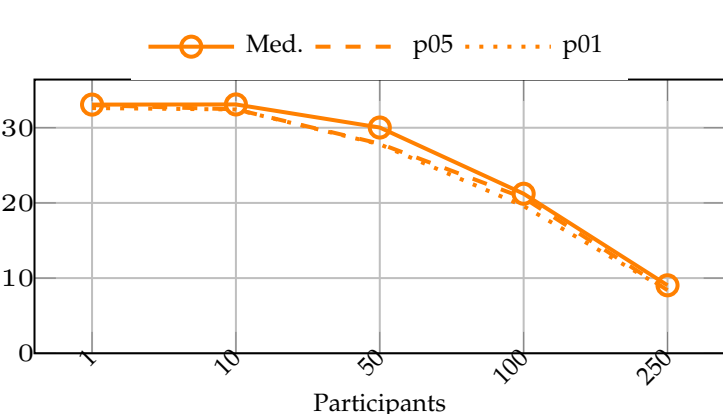
(c) Human-agent cooperation ③ with SP/BC in Overcooked: 29 LoC



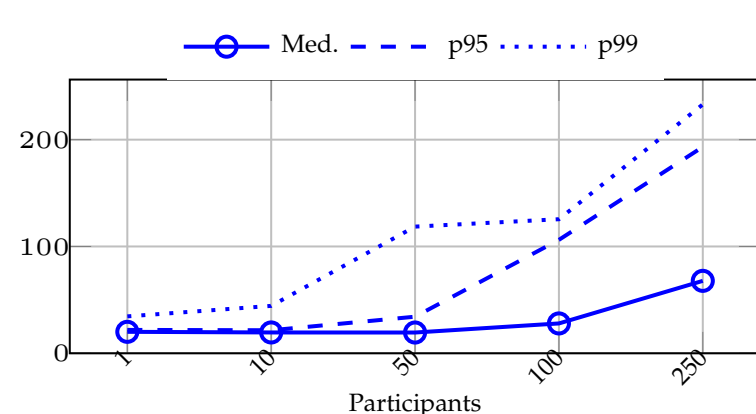
(d) Evaluative feedback ④ with TAMER in MountainCar: 31 LoC

Figure 1: Existing experiments reproduced with SHARPIE across interaction paradigms at low cost in terms of lines of code (LoC).

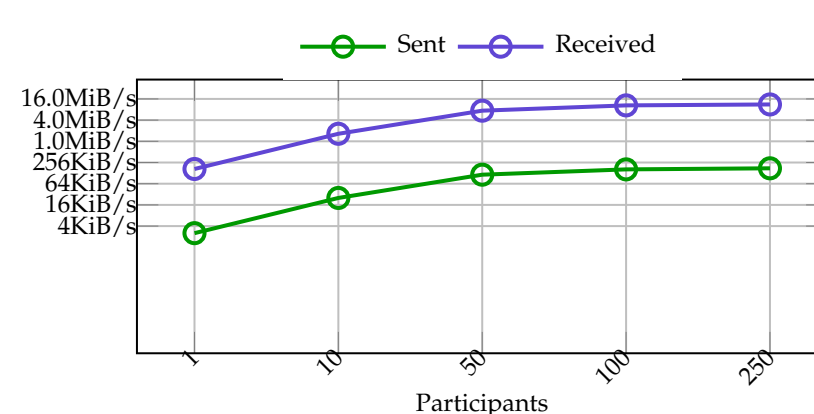
3. How does SHARPIE perform and scale?



(a) FPS

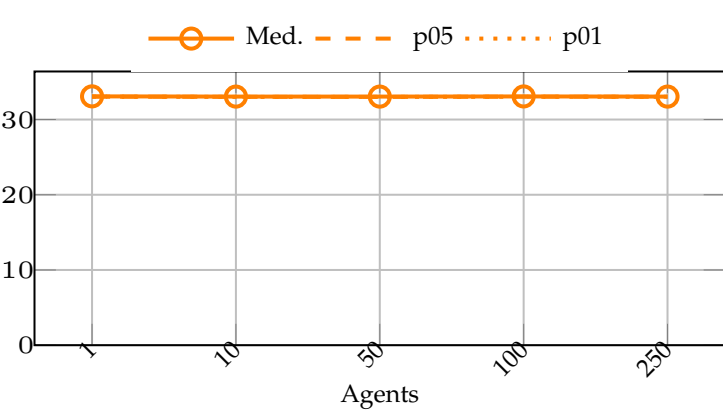


(b) RTT

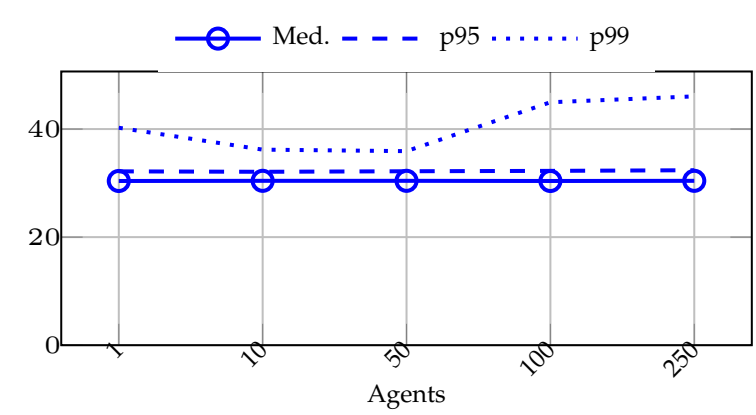


(c) Throughput

Figure 3: SHARPIE scales up to 50-100 simultaneous participants in a single real-time environment. Target FPS=30.0.



(a) FPS

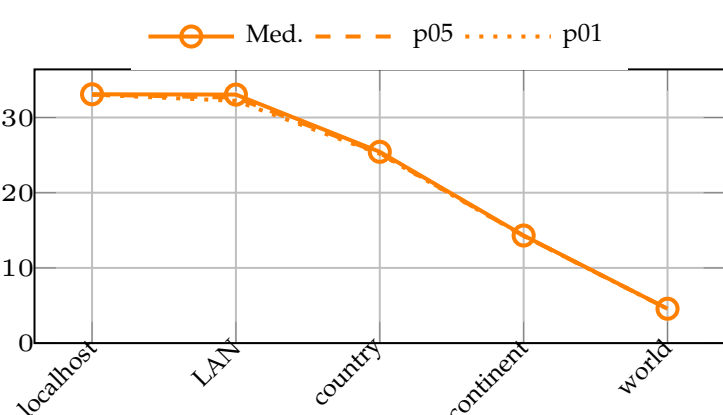


(b) RTT

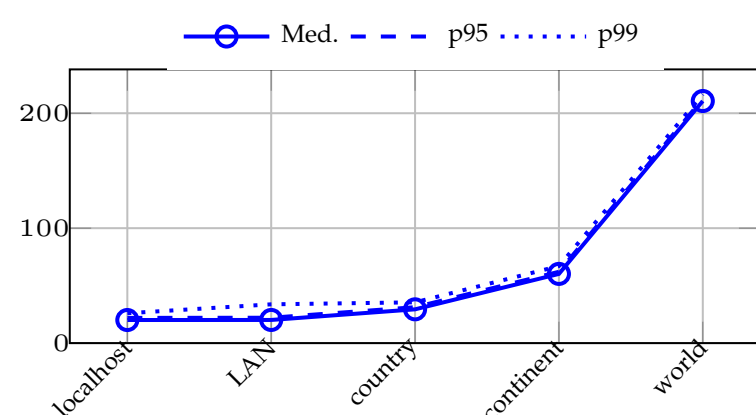


(c) Throughput

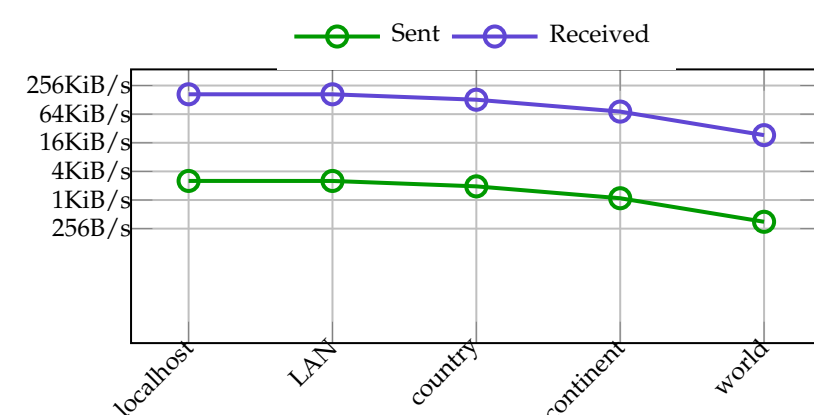
Figure 4: RL agents come with minimal overhead. Results shown for no-op agents in a dummy environment. Target FPS=30.0.



(a) FPS

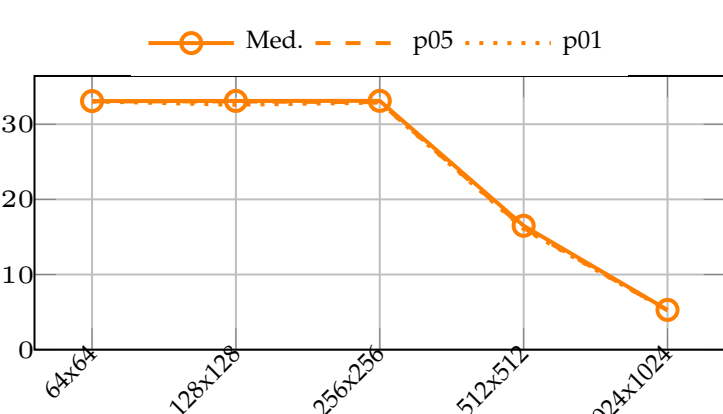


(b) RTT

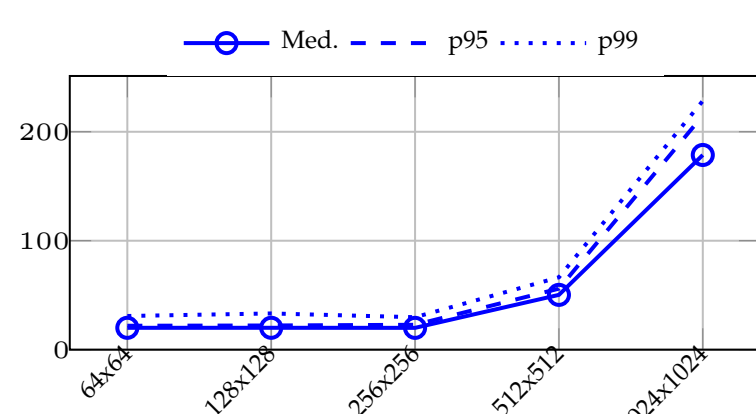


(c) Throughput

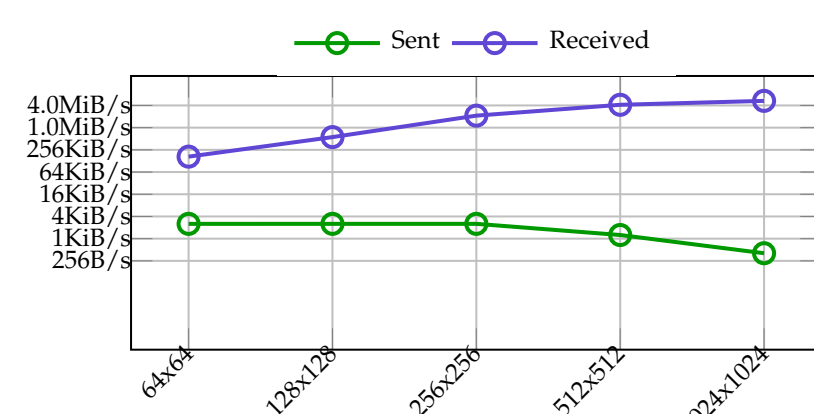
Figure 5: Real-time single-environment experiments can be conducted within a single country. Target FPS=30.0.



(a) FPS



(b) RTT



(c) Throughput

Figure 6: Maintains target FPS=30.0 for renders up to 256x256 for StarCraft II.

4. Features and Architecture

- a **Python-based web framework** following an extended multi-objective, multi-agent *Gymnasium API*.
- provides a **versatile wrapper** around and **UI** for popular RL environments, algorithms, and methodologies.
- comes with **utilities** for configuration, logging, and supports **crowdworker platform integration**.
- uses asynchronous participant input handling and web sockets (ws) for **server-side synchronisation** of environment state with participant and agent input.

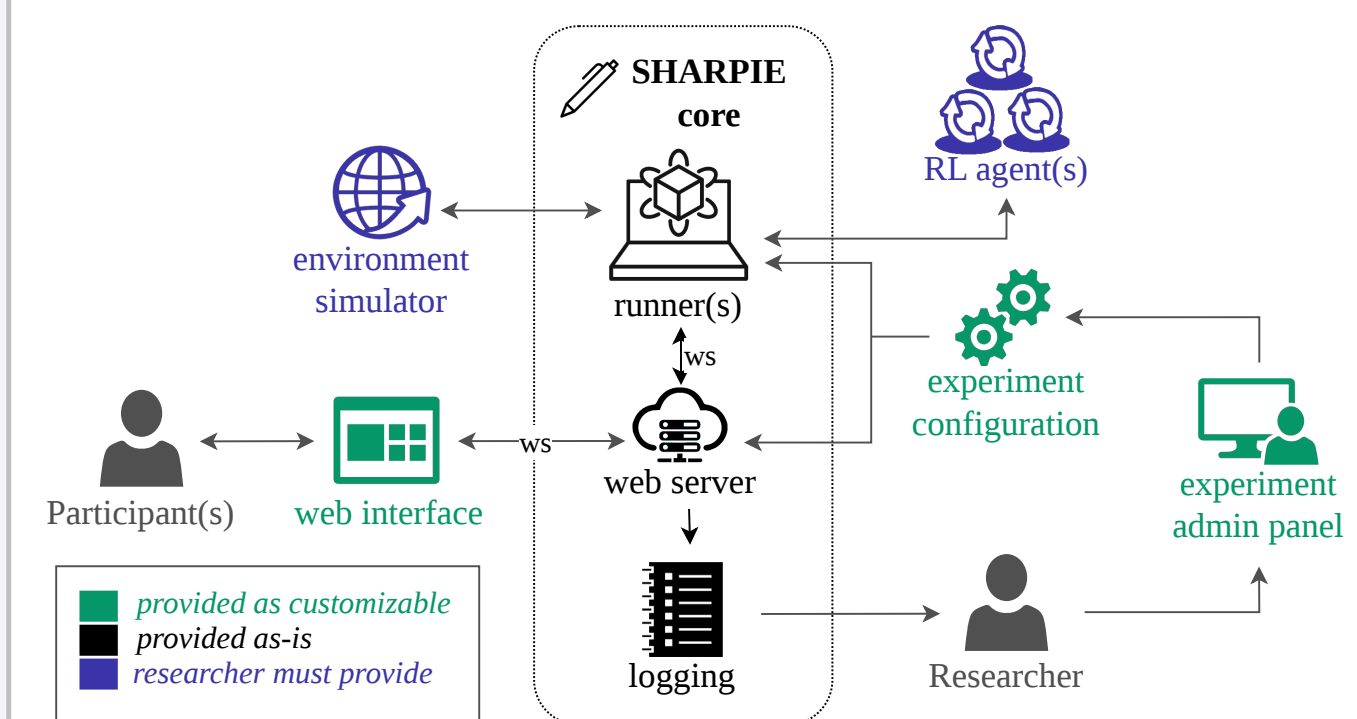


Figure 2: High-level SHARPIE architecture.

6. What will be next?

- a module for replaying experiments from logs
- support for high-frequency modalities: speech, video, ...
- additional tutorials, material and gallery examples
- better community integration

“to become a standard for controlled Hybrid Human-RL experimentation”

5. How to use SHARPIE?

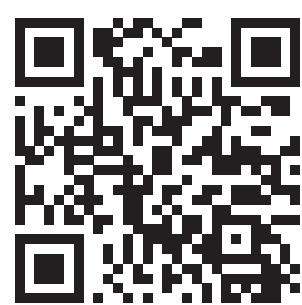
Paper



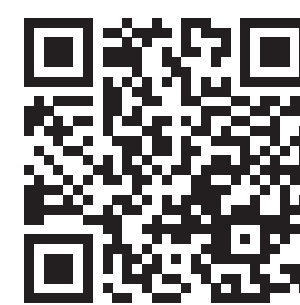
Code



Documentation



Hosted Demo*



*Use registration key “sharpie”

Acknowledgements

We thank Andreas Sauter and Mustafa Mert Çelikok for their input on an early design of SHARPIE.

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